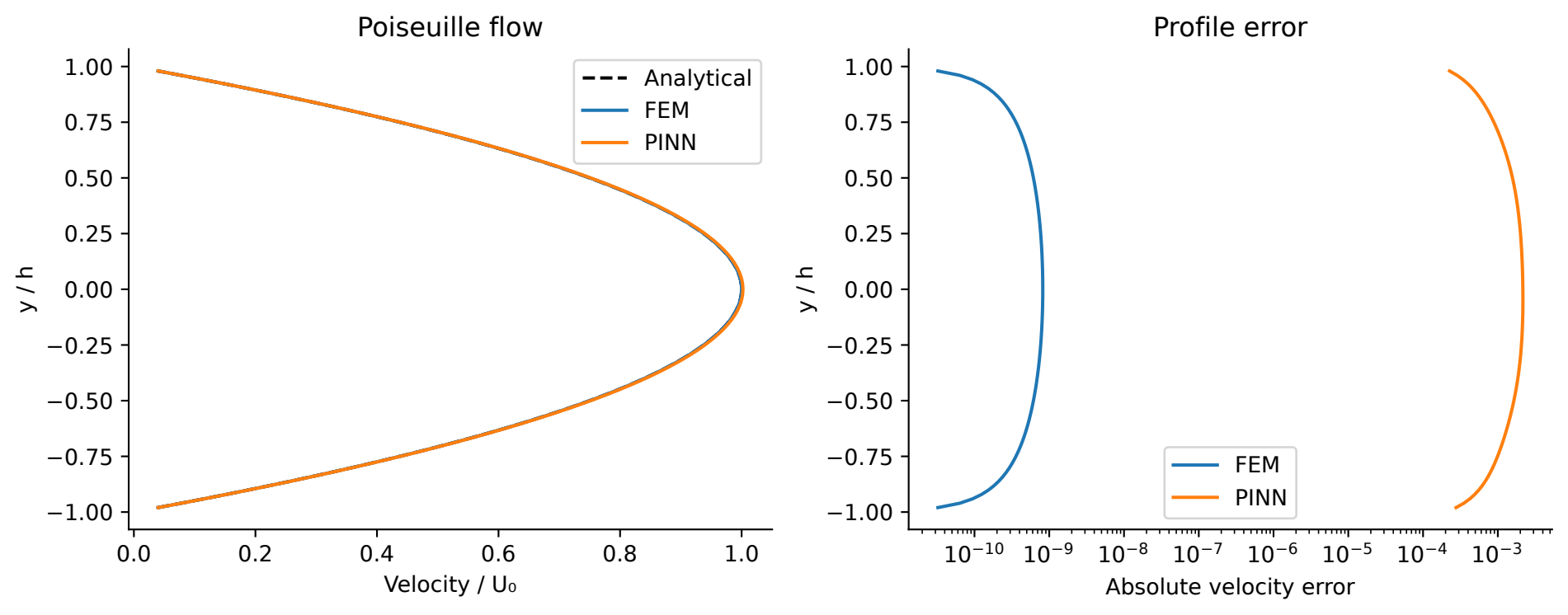
Poiseuille flow
Analytical
FEM
PINN
y / h
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0.0
0.2
0.4
0.6
0.8
1.0
Velocity / U₀
Profile error
FEM
PINN
10^{-10}
10^{-9}
10^{-8}
10^{-7}
10^{-6}
10^{-5}
10^{-4}
10^{-3}
Absolute velocity error